

RÚBEN ARAÚJO

QUANTITATIVE & RISK ANALYTICS

- Quantitative analysis of complex, high-dimensional datasets, from data curation and preprocessing through modeling and interpretation.
- Predictive modeling for classification and outcome forecasting under uncertainty, with explicit handling of noise, confounding factors, and class imbalance.
- Feature selection, dimensionality reduction, and multivariate analysis to extract stable, decision-relevant signals.
- Model evaluation and validation using performance and risk-aware metrics, including error trade-offs and robustness assessment.
- Time-dependent analysis and pattern recognition to capture evolving system behavior and regime changes.
- Translation of model outputs into risk-informed decision support, producing actionable insights rather than black-box outputs.

ANALYTICAL & DATA TOOLING

- Python for data analysis, modeling, and analytical automation.
- Data manipulation and numerical analysis using Pandas and NumPy.
- Statistical modeling and machine learning workflows with scikit-learn.
- Structured data querying and management using SQL (PostgreSQL).
- Exploratory analysis, reporting, and presentation using Excel (advanced), Power BI, and Tableau.

MARKET & RISK EXPOSURE

- Financial data analysis and strategy evaluation using quantitative methods.
- Risk-adjusted performance assessment, including drawdowns and return variability.
- Portfolio-level analysis, diversification principles, and exposure management.
- Familiarity with derivatives, leverage, and margin mechanics from a risk perspective.
- Key concepts: risk-adjusted returns, Sharpe and Sortino ratios, maximum drawdown, CAGR, volatility, exposure limits.

ENGINEERING HIGHLIGHTS

- Field-oriented problem solving in multidisciplinary engineering teams (mechanical, hydraulics, control systems, documentation, sales).
- On-site and international engineering experience, bridging design, manufacturing, and operational constraints, including leadership and coordination of technical teams.

EDUCATION

PhD in Biomedicine – NOVA Medical School (NMS)	2021 – 2026 Lisbon, Portugal
MSc in Biomedical Engineering – ISEL	2017 – 2019 Lisbon, Portugal
MSc in Mechanical Engineering (Maintenance & Production) – ISEL	2007 – 2011 Lisbon, Portugal
BSc in Mechanical Engineering – ISEL	2002 – 2007 Lisbon, Portugal

DATA & ANALYTICS SKILLS

Data Analysis	Statistical Modeling
Machine Learning	Multivariate Analysis
Time-Series Analysis	Risk Metrics
Python	SQL
Pandas / NumPy	Excel (Advanced)
Data Visualization	Power BI / Tableau

LANGUAGES

- Portuguese – Native
- English – C1 (IELTS + CAE)
- Norwegian – B1 (Aftenskolen)
- Spanish – B1 (Instituto Cervantes)
- French – Basic

AWARDS & SCHOLARSHIPS

FCT PhD Studentship	2021-2025
First in Class (PhD)	2023
Research Scholarship (ISEL)	2021
Merit Scholarship (Ministry of Education)	2021
First in Class (MSc Bio. Eng)	2020

ENGINEERING EXPERIENCE

CEO – Memory Particle (Portugal)	2017-2018
Lead Mechanical – Brøvig (Norway)	2016-2017
Life Cycle Engineer – MH Wirth (Norway)	2014-2015
Project Engineer – Aker Solutions (Norway)	2012-2014
Mechanical Design – Siemens (Portugal)	2011-2012

Note: Senior engineering experience in large-scale, multidisciplinary and multinational industrial projects, involving coordination across diverse technical teams, communication with multiple stakeholder groups, and interface with executive-level decision makers.

All supporting documentation can be consulted and downloaded @ <https://rubenaraujo.org/cv/>